Algoma UNIVERSITY

Assignment 3

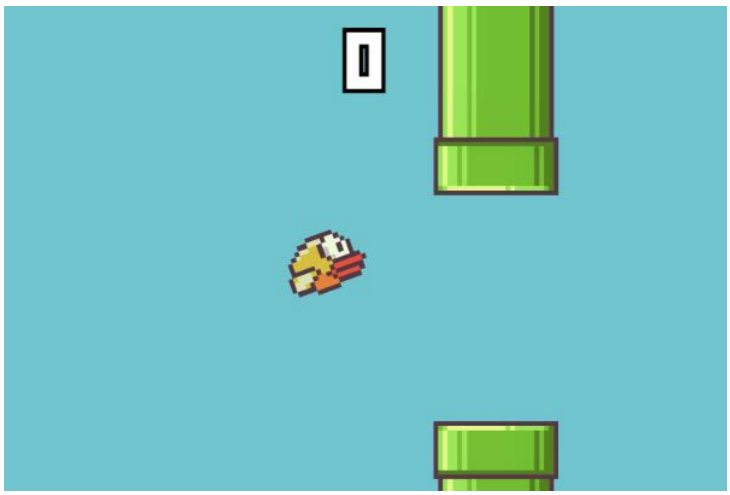
Total Marks: 9 marks

Course: Internet Tools
Topic: Game Application

In this assignment, you will extend the game we created in HTML, CSS and JavaScript **in Lecture 4 – Part 4 (https://www.youtube.com/watch?v=ZxVkYRQdxjs)**. The code for the game can be found on the Brightspace in Lecture 4 folder. I want you to extend the game.

Core Extensions:

1) **Dual Obstacles (Top and Bottom Pipes)**
    a. Create two obstacles (like the one I showed you in the class) moving together with a gap in between. Like one shown in this image. The gap should be enough to allow character to jump in between. Just to clarify, ***I don't want you to make something like an image below. This image is just to give you an idea of how one UP obstacle and one DOWN obstacle looks like. I want you to work on original game we created in Lecture 4, and extend it instead of writing code from scratch)***



2) **Horizontal Movement**
    a. Add **ArrowLeft** and **ArrowRight** to move the player horizontally.
    b. Prevent moving out of the game box.
3) **Scoring System**
    a. Print **Score** and increase **it** when player successfully passes an obstacle pair.
4) **Health**
    a. For each game play, only 2 chances are allowed. After 2 chances, the only way to start the game is to reload the page.

Deliverables:

- Record the demonstration as a short video (1 – 3 minutes) and submit it along with your code.
- Run the app in your PC.
- Show the game is working fine.
    - **Move left and right to show you have implemented functionality**
    - **Jump to pass obstacles**
    - **Score should be increased in case you jump correctly.**
    - **Die 1 time to show you can start game from the alert. Die 2nd time and show game cannot be started from alert and you have to reload the page.**
- Upload the code on GitHub.
- Provide the YouTube video link and GitHub code in the text file.

Note: Failure to follow the instructions will result in ZERO marks.